

Inductance Spreadsheet

Don Peterson 5 Dec 2010
someonesdad1@gmail.com

This package provides a spreadsheet to perform low frequency inductance calculations for various configurations.

The **inductance.zip** file contains the following files:

inductance.odt	This file
inductance.pdf	The spreadsheet converted to PDF form
inductance.ods	The inductance spreadsheet
inductance1.ods	Simpler version of the inductance spreadsheet
grover.py	Python module for the inductance.ods spreadsheet

I wrote the spreadsheet using Open Office (OO) 3.2.1, so I recommend you use that version of OO for best results. Things will probably work fine with earlier versions because there's nothing terribly sophisticated in the spreadsheet or macros, but I haven't tested it with any earlier versions.

If you do not have Open Office version 3, you will not be able view or use the spreadsheet. However, you can see what the spreadsheet looks like by viewing the PDF file.

The drawings in the spreadsheet are embedded OO Draw drawings, so you can modify them if you wish. The input and output cells' formatting is controlled by styles. There are a number of different styles defined should you wish to change how numbers are displayed. Press F11 in Calc to bring up the list of styles so that you can apply or edit them.

The two spreadsheet versions

There are two spreadsheets supplied. **inductance1.ods** is the simpler version in that it has fewer inductance formulas. It is self-contained and uses OO macros to perform its calculations.

The other spreadsheet, **inductance.ods**, uses a python module **grover.py** to perform a number of its calculations. You will have to copy this python module to a specific directory in order for the Open Office spreadsheet to be able to use it. This directory depends on your platform; for Windows, it will be something like

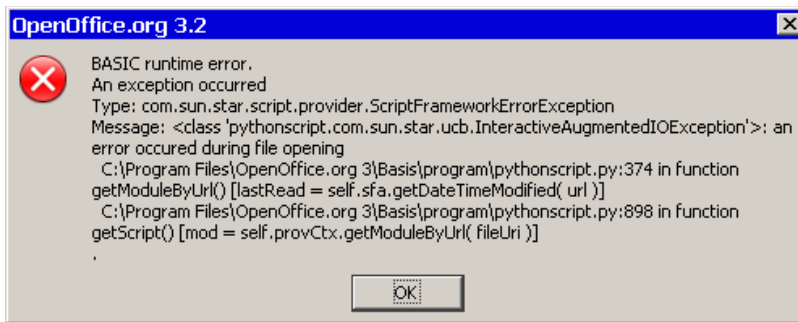
c:\Documents and Settings\<user>\Application Data\OpenOffice.org\3\user\Scripts\python

and for UNIX-type systems it will be something like

~/.openoffice.org/3/user/Scripts/python

Note you will have to make the **python** directory if it does not already exist. Then copy the **grover.py** script to this directory.

This python module will be recognized as a macro and you should be warned about running untrusted macros, as discussed in the next section. If you try to open the spreadsheet without this python module being available, you'll get an error message like the following:

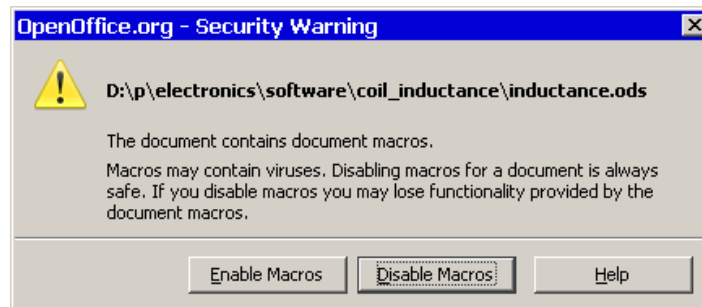


You'll still be able to use formulas that don't depend on the python module's functions, but you'll get an error message and the Basic macro editor will open when you try to update cells that do depend on this module. You can safely ignore the error messages and dismiss them, but the relevant inductance cells will have a result of zero.

Loading the spreadsheet the first time

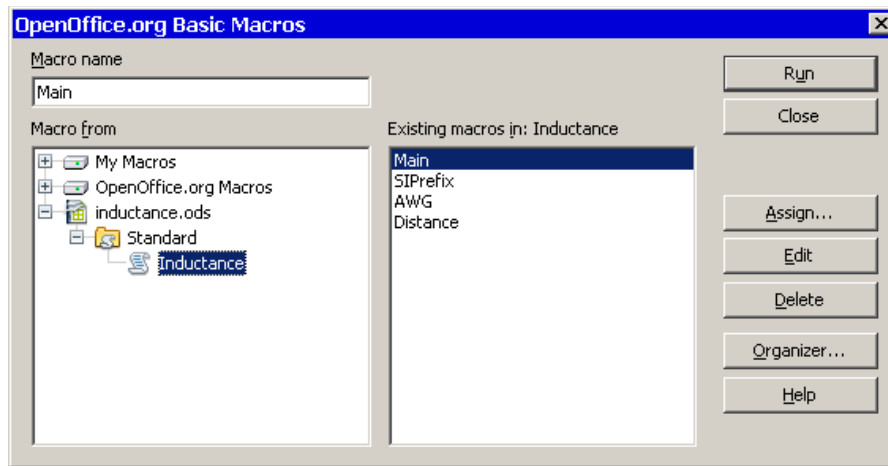
The first time you load the spreadsheet, you'll probably get a warning that the spreadsheet contains macros and you're asked whether you want to enable them or not. If you don't get this warning, your macro protection settings are weak and, unless you're sure you know what you're doing and why, you should go into **Tools->Options** and adjust them for more security under the **OpenOffice.org->Security** tab (click on the **Macro Security...** button).

This is OO's way of protecting you from malicious code. Here's the dialog you'll see (the file location will be different):



To be safe, click on "Disable Macros" and open the spreadsheet. Your first step should be to verify that the macros that come with the spreadsheet are safe to execute. To do this, use **Tools->Macros->Organize Macros->OpenOffice.org Basic...**

You'll see the following dialog:



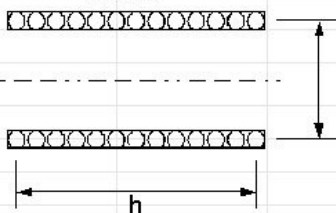
Click on the Main macro, then the **Edit** button. This will open up the BASIC editor/IDE and allow you to inspect the code that makes up the macros. You should read through the code and satisfy yourself that the code is not trying to do anything malicious to your computer. It's beyond the scope of this document to define for you what "malicious" means in this context. If you're not sure, then don't use the spreadsheet. If you're using the spreadsheet that needs the python module, then you should also check that python module for malicious behavior.

If, however, you feel that the macros are safe, then you can exit the editor and the spreadsheet and start the spreadsheet again, this time enabling the macros. Then the spreadsheet is ready for use.

If you wish, you can avoid the dialog to enable the macros on opening the spreadsheet by using **Tools->Options->Security** and clicking on **Macro Security...**. Then define a directory to store trusted spreadsheets and put the inductance spreadsheet there.

Basic Usage

Once the macros are enabled, the spreadsheet is ready to use. You should see something like the following when the spreadsheet is started (this picture may not represent the latest version of the spreadsheet, but it will give you the general idea):

	A	B	C	D	E	F	G	H
1	Approximate inductances of coils and wires							
2	All formulas are low frequency formulas except where noted. The sheet is protected (but with no							
3	password) to avoid inadvertent changes. You can modify those numbers and units with a pale							
4	yellow background. Negative inductances in red indicate an input quantity out of bounds.							
5								
6	The cell to the right of a variable indicates its unit of measurement. You can specify the length							
7	variables with SI prefixes and use m, in, or ft. If you specify AWG, the equivalent American Wire							
8	Gauge diameter is used; use -1 for 2/0, -2 for 3/0, and -3 for 4/0. The length unit is case-							
9	insensitive (this is because of Open Office's quirky and broken BASIC macro language).							
10	Similarly, inductance is in H with an optional SI prefix. Temperature units can be C (Centigrade),							
11	F (Fahrenheit), K (Kelvin), or R (Rankine). Resistance units are ohms with optional SI prefix.							
12								
13	Wire resistance				Convert AWG sizes		16	AWG
14					Diameter	d	1.29	mm
15					Diameter	d	0.0508	in
16					Material		copper	
17					Temperature	T	100	C
18						h	1	m
19						R ₂₀	13.17	mohm
20					Resistance at indicated temperature	R _T	17.32	mohm
21								
22	Single layer air-core solenoid				Coil diameter	D	1.1	in
23	Coils are close-packed				Coil length	h	10	cm
24	Wheeler's 1928 formula, assumes h/D > 0.4				Number of turns	n	200	
25	n turns				Wheeler's 1928 formula	L	273.77	uH
26					h/D		3.58	
27					Wheeler's 1982 formula	L _s	274.83	uH
28					Wire diameter	d	0.1	mm
29					Grover	L	277.15	uH
30								
31								
32								

The spreadsheet is protected against modifications except in the cells with the yellow background. If you see gray cells with white cells where the yellow cells should be, you probably have Value Highlighting on; press Ctrl-F8 to turn it off. You can, of course, turn the protection off if you want to change any aspects of the spreadsheet. After you're finished, I recommend you turn the protection back on, as it's just too easy to accidentally delete a formula in a cell by overwriting. If you plan to modify the spreadsheet, I recommend you make a backup copy before making your modifications or first check it into a version control system.

The spreadsheet is easy to use -- scroll to the picture showing the case you want to calculate, then enter appropriate numbers and units into the yellow cells and get the calculated inductance. The formula used for each inductance is given to the right of the cell containing the inductance. For formulas that are too complicated to display or that are in the python module, you are given a reference where to look for more detail.

As indicated in the instructions at the top of the spreadsheet, you can use length units of m, in (inches), and ft (feet). These units can have an SI prefix if you wish (use u for 'micro'). You can also use "AWG" as a unit and the equivalent American Wire Gauge wire size will be used.

These formulas represent a lot of work by a number of people over the years. The formulas can produce good output for good input. Therein lies the rub: **it's your responsibility to determine whether the input is good**. In general, this would require you to consult the references given and,

in turn, consult the original papers that presented the formulas and use the information in that paper to limit the inputs to those that the author deemed proper for his formula. But to do this is a lot of work, even assuming you have access to a large library that can get you the needed references. It would be no significant programming effort to include code to check the inputs and make an estimate of the reliability of the answer once one had access to the original papers or suitable references. But this is unlikely to be done unless someone is willing to provide me with the papers or do the work themselves.

Instead, here's how I view the spreadsheet: it's a tool to help you make estimates of a resulting inductance to one or two significant figures. Some of the formulas are capable of more precise predictions, but it will take study on your part to determine which ones. One aid the spreadsheet provides is that inductance calculations that use out-of-bounds inputs will display a negative inductance in red. This will only be done by those inductances that are calculated by the python module; you'll have to refer to the Basic macro code to determine which those are.

Also, remember that these are formulas used to make low-frequency inductance predictions except where noted. I interpret this to mean that the formulas predict approximately what inductance I will measure with my LCR meter set to 100 Hz or 1 kHz test frequency.

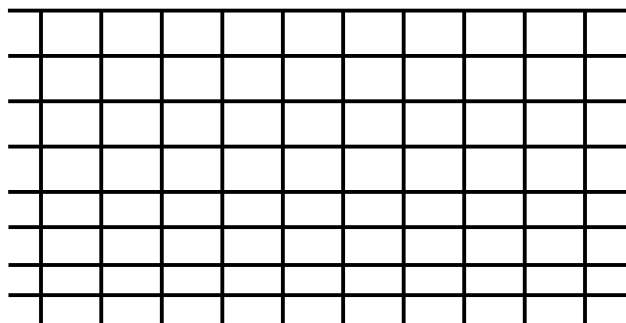
Example

Let's illustrate the use of the spreadsheet to calculate an example. My choice of an example is perhaps of little practical use, but it illustrates the use of the spreadsheet.

As I look outside my window while I'm typing this, I see a some wire fencing that we use to keep our animals in our yard. The chunk of fence I'm looking at is 21 m long, about 1 m tall, and has vertical strands of wire every 150 mm and horizontal strands that are separated by from about 90 to 150 mm (the horizontal strands are closer at the "bottom" of the fence than at the top (this helps keep smaller animals from getting out, but also conserves wire). The fence is supported by metal fence posts 2.4 m apart. If I were to run a 16 gauge wire to each end of this fencing, then measure the inductance, what value would I measure? Considering that it's a cold winter day in December with snow on the ground, I'm not too inclined to go out and make a measurement. But we can make an estimate using the spreadsheet. Of course, I will have to go out in the cold and make a measurement, as no one (including me) would be interested in what the spreadsheet predicted without comparing it to an actually-measured value. Why didn't I write this material on a warm summer day? ☺

The problem is one that probably most people (including me) wouldn't be able to estimate the inductance within an order of magnitude. The spreadsheet can help you estimate the order of magnitude.

Here's roughly what the fence looks like:



There are 9 horizontal wires; the actual spacings of the horizontal wires from the bottom wire in mm are 78, 82, 105, 110, 122, 141, 147, and 180. The top and bottom wires are 1/8" in diameter; the other wires are 0.095 inches in diameter.

The question is how to model this inductance using the elementary inductances given in the spreadsheet. Here are some possible models.

One configuration we could use would be 9 wires in parallel, resulting in 9 independent inductors in parallel. Since I will use two 15 m extension cords with 16 gauge wires to connect to the fence ends, we'd add the inductance of them in too. A 21 m long wire of 0.095" in diameter has an inductance of 35 μH . Nine of these in parallel are $35/9 \mu\text{H}$ or 4 μH . Each extension cord adds 26 μH . Thus, the measured inductance by this model would be about **56 μH** .

Another model would be pairs of parallel wires. The paired inductances for the measured separations are, respectively in μH , 37, 37, 40, 40, 41, 42, 42, and 44. These can be put in parallel and result in 5 μH . This model thus predicts **57 μH** .

Another model would take the mean size of a wire rectangle, 120 by 150 mm, and consider the fence as 8 strings of those rectangles with $21000/150 = 140$ rectangles in each string. The rectangles add in series, then each string adds in parallel. Each rectangle is about 0.5 μH . Each string is thus 70 μH and 8 of these in parallel results in about 9 μH . Add in the extension cords to get the result **61 μH** .

One can see that the fence itself is contributing about 10% of the overall measured inductance, about in the 3 to 9 μH region. Of course, these models don't truly model the inductance of the fence, as they are too simplified. But they serve to give an estimate; things will also be complicated by the fact that the fence is in electrical contact with the ground, so that supplies another current path. Because above I said the program will help to estimate the order of magnitude of the inductance, I would guess that the total measured inductance will be in the 10 to 100 μH range. I don't have enough experience with inductance to make a better guess than that. The spreadsheet also predicts the inductance of the extension cords will be 52 μH , so the estimated inductance of the fence should be within an order of magnitude of about 5 μH .

One extension cord measured 44' long. I measured the inductance of its ground wire, necessarily forming the cord in a loop around the room. The inductance was 18 μH . The spreadsheet predicts 23 μH for a round loop of wire with a circumference equal to the length of the extension cord, so the numbers are reasonable; I'll trust the 26 μH for an extension cord when it is straight.

The prior material was all written before I went outside to make a measurement.

Details

In the discussion below, when formulas are stated as coming from [grover], the spreadsheet implementation uses a python script that provides the calculations. The python script implements some of the formulas from [grover] in that the tables given in [grover] are reproduced and the functions interpolate in these tables.

Single layer air-core solenoid

One of the most popular formulas is Wheeler's 1928 formula because it is such a simple form. Note it assumes the length/diameter ratio of the coil is > 0.4 . Wheeler updated this formula in a 1982 paper and stated it was accurate to around 0.1%; note it applies to coils of any length/diameter ratio.

The cell labeled Grover is based on formula 118 in [grover]. It includes Nagaoka's correction and Rosa's correction for wire size. From what I've read, the indications are that it is capable of predicting inductance to within a few parts per million and, as such, could be used to design laboratory standard inductances. The typical hobbyist will not be able to use this formula's precision because of the inability to measure linear dimensions to 5 or 6 significant figures. However, it's still nice to have this "industrial strength" formula available to compare the Wheeler formula predictions to.

Note that the Grover result can be used for helices of pitch larger than close-packed windings. I can't give you any guidance on how accurate the formula is as the pitch increases, but I suspect it's adequate for most coils used in practice.

Here are comparisons to example 46 in [grover], which is for a coil with mean radius 27.0862 cm (diameter of 54.1724 cm), length of 30.5510 cm, bare wire diameter of 0.0634 cm, and 440 turns:

	L, H	%dev from Grover
Wheeler 1928	0.10208522	0.39
Wheeler 1982	0.10183953	0.15
Grover	0.10169011	--

Multiple layer air-core solenoid

The Wheeler formula is another formula Wheeler developed for coils with multiple layers. This is compared to the Brooks inductance formula [brooks], which is said to be within 3% for all multiple-layer coils. Both of these formulas should be compared to the two Grover formulas, "Grover disk" and "Grover thin". These two formulas are from [grover], formulas 100 (page 110) and 99 (page 105), respectively. The thin formula is for coils with small radial thickness and the disk formula is for coils that are thick in the radial direction compared to length in the axial direction (i.e., they look like a disk). They are both appropriate for coils of intermediate shape and provide a check on the calculations. Both formulas are proportional to the coil radius multiplied by the square of the number of turns. Formula 99 also includes a factor of the diameter/length.

Wire information

At the top of the spreadsheet is a set of cells for converting American Wire Gauge sizes to physical dimensions (I've put two output cells so the output can be viewed in mm and inches at the same time, but you can change them to units of your choice). This section of the spreadsheet also calculates the resistance of a length of round wire of that diameter. Select the metal to use (choices are listed to the right) and the ambient temperature; the spreadsheet will calculate the wire's DC resistance at 20 °C and the indicated temperature. Note that this will only work over the temperature range where the change in resistance with temperature is linear. For larger temperature ranges, consult tables such as the Smithsonian Physical Tables or the American Society of Metals *Metals Handbook*, various volumes. You can also use different temperature units (either in upper or lower case) as stated by the comments at the top of the spreadsheet.

The values in the table for the resistivity temperature coefficients came from the CRC Handbook of Chemistry & Physics, 59th ed., 1978. Warning: you will find a variety of values for the temperature coefficients of resistivity on the web. Part of this is due to the variation due to temper, alloying, etc. of the metal; however, part is due to errors. The careful experimenter will in fact simply measure the property for himself with the material at hand.

The number you enter for the AWG number can be any number; the conversion will still be done, even if it's not a legitimate AWG number. While you'd have to be a raving lunatic to come up with a scale like the AWG scale, since it's a scale based on geometric progressions, you can express large and small diameters with modest AWG numbers. For example, the range of 155 to -176 AWG spans the distance range of the radius of the copper atom to the radius of the Earth.

For folks who want to just enter wire diameters as a linear dimension such as mm directly, you have two choices. If you want to leave the AWG conversion intact, you can use Calc's goal seeking functionality to find an equivalent AWG size for a given diameter. Use the d cell just below the AWG cell (because that's the cell used in the resistance formulas below) as the target cell. See the next section for details on goal seeking.

If you know you don't want to keep the AWG conversion, you can just change the first d cell to be a number instead of the formula it currently contains. I suggest you change the cell's style to indicate that it's a value to be changed (otherwise you won't be able to change the cell's value if you protect the sheet).

Goal Seeking

You can use the goal seeking feature of Calc to find one of the yellow-background parameters given a certain inductance. Here's an example.

Suppose I have some 16 gauge wire and I want a round loop of this wire with an inductance of 5 μH . How big do I need to make the loop?

Enter 16 in the cell for d and AWG in the cell for d's units.

Click on **Tools->Goal seek** and select the cell with the loop inductance as the target cell. Enter 5 as the target value. Enter the diameter D cell as the cell to be varied and click OK. The answer will come up as 1090.13 mm.

Here's another example using the wire conversion section. Suppose you want to perform a wire calculation for a wire that is 0.045 inches in diameter. In Goal Seek, choose the second d cell as the target (make sure the units are in inches). Set the AWG cell to be the one to be varied. You'll find the equivalent AWG size will be 17.05.

Here's a final example that illustrates a typical problem. Suppose you have a 20 foot long piece of 3/4 inch diameter round brass bar stock. You want to estimate its longitudinal resistance at 100 °C. Enter "in" into the second d cell's unit cell, then perform a goal seek to find the equivalent AWG size to make the second d cell 0.75 inches (it will be an AWG number of -7.22). Enter 20 ft for the length of wire, brass for the material, and 100 C for the temperature T. You'll get a resistance of 1.737 m Ω at 100 °C.

You can measure such resistances if you have a constant current source of, say, 1 to 10 A. Measure the voltage drop across the conductor as it is conducting the constant current and use Ohm's Law to calculate the resistance. Make sure that the voltmeter's connections are inside of the connections to the current source so that the contact resistance of the current connections isn't measured (this makes it a Kelvin 4-wire measurement).

grover.py

This section discusses the **grover.py** script. My intention in writing this script was to reproduce some of the inductance calculations given in [grover]. One's first inclination in developing such calculations is to get the references Grover refers to and program the formulas given in those papers. Since I don't have easy access to a large library that might have such references, it was easier to type in the tables Grover gives and interpolate in them, much as a normal user of the book would do.

One of the problems of this method, however, is the risk of errors in the table values, both in the original document (i.e., [grover]) and because of errors in my transcription. I tried to minimize such errors by a variety of methods. Where formulas were available, they were used to validate the table data. The table values also were plotted; errors were visible due to the lack of continuity in the functions (this found 4 or 5 errors in various tables). Finally, some table data were printed out in a form that made manual checking with the values in [grover] easier.

Validation

One of the problems with using formulas is it's so easy to make a mistake, such as writing down the formula wrong, using the wrong units, or making a coding mistake. Because of these problems, it's handy to validate the formulas' output with values that make it easy to figure out the inductance by hand. The following table contains the calculated inductances for the indicated parameters; you can verify these values by substituting them in the formula to the right of the inductance cell. Note that there's no claim that the output makes physical sense -- this is just intended to validate that the

formula is being calculated correctly.

Single layer air-core solenoid	$D = 1 \text{ m}$, $h = 1 \text{ m}$, $n = 10$, $L = 68.07 \mu\text{H}$ (Wheeler 1928), $L_s = 67.98 \mu\text{H}$ (Wheeler 1982)
Multiple layer air-core solenoid	$D = 1 \text{ m}$, $h = 1 \text{ m}$, $t = 1 \text{ m}$, $n = 10$, $L = 35.79 \mu\text{H}$ (Wheeler), $L = 37.12$ (Brooks)
Straight round wire	$d = 1 \text{ mm}$, $h = 1 \text{ m}$, $L = 1.23 \mu\text{H}$
Loop of round wire	$D = 1 \text{ m}$, $d = 1 \text{ mm}$, $n = 1$, $L = 4.55 \mu\text{H}$
Rectangle of round wire	$w = 1 \text{ m}$, $h = 1 \text{ m}$, $d = 1 \text{ mm}$, $n = 1$, $L = 5.66 \mu\text{H}$
Polygons of round wire	$d = 1 \text{ mm}$, $s = 1 \text{ m}$, triangle $L = 3.87 \mu\text{H}$, square $L = 5.66 \mu\text{H}$, pentagon $L = 7.44 \mu\text{H}$, hexagon $L = 9.00 \mu\text{H}$, octagon $L = 12.10 \mu\text{H}$
General planar loop of round wire	$d = 1 \text{ mm}$, $p = 4 \text{ m}$, $A = 1 \text{ m}^2$, $L = 5.73 \mu\text{H}$
Planar spiral	$D = 1 \text{ m}$, $t = 10 \text{ mm}$, $n = 10$, $L = 239.48 \mu\text{H}$
Two parallel wires	$D = 10 \text{ mm}$, $d = 1 \text{ mm}$, $h = 1 \text{ m}$, $L_{LF} = 1.30 \mu\text{H}$, $L_{HF} = 1.20 \mu\text{H}$
Wire parallel to ground plane	$d = 1 \text{ mm}$, $D = 10 \text{ mm}$, $h = 1 \text{ m}$, $L_{LF} = 0.7390 \mu\text{H}$, $L_{HF} = 0.7377 \mu\text{H}$
Parallel rectangular traces	$w = 1 \text{ cm}$, $D = 1 \text{ mm}$, $h = 1 \text{ m}$, $L = 0.1257 \mu\text{H}$
Wide trace over ground plane	$w = 1 \text{ cm}$, $D = 1 \text{ mm}$, $h = 1 \text{ m}$, $L = 0.1257 \mu\text{H}$
Coplanar traces	$w = 1 \text{ mm}$, $D = 1 \text{ cm}$, $h = 1 \text{ m}$, $L = 1.20 \mu\text{H}$
Isoceles triangle loop of wire	$d = 1 \text{ mm}$, $b = 1 \text{ m}$, $s = 1 \text{ m}$, $n = 1$, $L = 3.72 \mu\text{H}$
Coaxial cable	$d = 1 \text{ mm}$, $D = 10 \text{ mm}$, $h = 1 \text{ m}$, $\mu_d = 1$, $L = 0.4605 \mu\text{H}$
Toroidal inductor	$d = 1 \text{ mm}$, $D_i = 10 \text{ mm}$, $D_o = 100 \text{ mm}$, $h = 1 \text{ m}$, $n = 100$, $\mu = 1$, $\rho r = 1$, $\alpha = 1.05$, $L = 4605.17 \mu\text{H}$, $w = 104.60 \text{ m}$, $R = 2.296$

The python script contains its own self-tests. Run it as a script to run the self tests. Note most of the tests duplicate examples given in [grover].

For the spreadsheet, using the examples in [grover] is a good way of checking the formulas. The comments in the [grover.py](#) script indicate which tests were checked with and the spreadsheet cells that use the relevant functions should reproduce Grover's results to within one count or so of the last significant figure.

An actual coil

I wrapped 46 turns of insulated 16 gauge stranded wire (0.1 inch outside diameter) around a 1.1 inch diameter cardboard tube to make a single layer air-core coil. This makes the coil diameter 1.2 inches. The length was 120 mm. Note that there is thus a separation between the windings and Wheeler's formula is for tightly-packed, close coils. However, we're just looking for an estimate and this formula will get us in the ball park -- Wheeler's formula gives 14.51 μH . A calculation using [moshier] will give 14.45 μH . Grover's formula gives 14.40 μH .

I measured the inductance of the coil with a B&K 886 LCR meter with the 4 wire Kelvin clips. The clips have a baseline inductance of 0.17 μH , which I subtracted from each reading. The measured inductance of the coil was

f, kHz	L, μH	Q	L with steel
0.1	14.58	0.63	92.88
1	14.53	1.16	37.75
10	14.54	1.93	18.78
100	14.36	3.77	11.98

The instrument measures inductance at different frequencies. The Q is the ratio of the coil's reactance to its DC resistance. The L with steel is the inductance with an 18" long chunk of 3/4" mild steel round bar stock I had on hand (I just wanted to see the effect of a ferromagnetic core).

The 1928 Wheeler formula made an excellent estimate (the 1982 formula gives a value 0.4% higher). If you enter the same values into the multiple layer air-core solenoid section and use $t = 0.05$ inches, you'll get 13.1 μH -- a value still in the ball park. Such cross-checking can help give you confidence in the reasonableness of the results.

References

There is a rich history of inductance calculations. The US NBS (National Bureau of Standards, now called NIST) had a number of people devote significant effort in the first half of the 20th century to developing equations to predict inductance (and illustrious 19th century names like Maxwell, Kirchoff, and Kelvin worked on these problems too). Part of this was no doubt due to the fact that a good formula let one construct an inductance standard if the linear dimensions were accurately measured. For example, both [grover] (page 149) and [rosa] (page 129) give the example of a coil constructed on an accurately-measured cylinder of marble with length and radius measured to 6 significant figures; they then proceed to calculate the inductance to 7 significant figures. However, they don't state what the actual measured value of this coil was.

If you enjoy programming, you may find yourself falling down the rabbit-hole of getting some of these references and thinking you can easily program them up yourself. Feel free to do so, but others have done a fair bit of work, so I suggest you first look at what they've done and see if you can leverage it. In particular, I refer you to [moshier], [weaver], and [knight].

- [cvel] The Clemson University Vehicular Electronics Laboratory Inductance Calculator,
http://www.cvel.clemson.edu/emc/calculators/Inductance_Calculator/index.html
- [brooks] M. Brooks and H. Turner, "Inductance of Coils", 1912 (can be retrieved from Google books).
- [grover] F. Grover, *Inductance Calculations: Working Formulas and Tables*, Dover Publications (reprint of 1946 book). This is usually recognized as the standard work on the subject. A tidbit is that Grover was born in 1876 (three years before Maxwell died) and died in 1973, so he lived almost 100 years. A disadvantage of this book for use with computers is that most of the formulas are used with tables that reduce the amount of calculation needed. This made sense before computers and calculators were available, but it means that you won't have the basic formulas that you could program into a computer unless you consult the references Grover gives.
- [knight] D. Knight, *Rosa's mutual inductance correction for the round-wire solenoid*, 3 April 2010, available from
http://www.g3ynh.info/zdocs/magnetics/appendix/Rosa/Rosa_km.pdf. I recommend you explore Knight's website, as he has quite a bit of material on inductance calculations.
- [moshier] Steven Moshier, COIL02 program, <http://www.moshier.net/coildoc.html>. Moshier's program is written in C and comes with a UNIX-style makefile that will compile the program with gcc. He references the NBS papers that give the formulas he uses. He provides the solutions for a number of different coil configurations.
- [rosa] E.B. Rosa and F.W. Grover, *Formulas and Tables for the Calculation of Mutual and Self-inductance*, NBS publication 169, 1948. Available from
<http://www.g3ynh.info/zdocs/refs/NBS/Sci169noerr.pdf>.
- [thompson] M. Thompson, *Inductance Calculation Techniques*, Part 2, 1999,
<http://www.thompsonrd.com/induct2.pdf>. Thompson's paper covers a number

of practical cases.

[weaver]

R. Weaver, Investigation of Rosa's round-wire mutual inductance correction for the solenoid inductor, see <http://electronbunker.sasktelwebsite.net/CalcMethods2c.html> and related pages. Weaver's website has some Open Office spreadsheets you can download to perform inductance calculations with. One is <http://electronbunker.sasktelwebsite.net/DL/InductCompare01.ods>. Also check his references page <http://electronbunker.sasktelwebsite.net/CalcMethodsRef.html>, because it contains copies of some of the original papers. Weaver also has a nice paper on the evaluation of elliptic integrals from their series expressions, as they converge quickly. If you don't have access to a library that computes these functions for you (like, e.g., SciPy), then you should take a look at Weaver's functions (they're very easy to translate into python).

This page on the site gives Wheeler's updated (1982) formula <http://electronbunker.sasktelwebsite.net/CalcMethods3b.html> as equation 30. [knight] gives a slightly optimized form in his equation 6.9 in section 3.6b.

[wheeler]

H. Wheeler, *Simple Inductance Formulas for Radio Coils*, Proc. I.R.E., vol. 16, pp. 1398-1400, Oct. 1928. This is the source of the widely-used Wheeler formula, but note the requirement that the coil not be short. Wheeler updated this formula in 1982: H., Wheeler, *Inductance Formulas for Circular and Square Coils*, Proceedings of the IEEE, Vol. 70, No. 12, December 1982, pp. 1449-1450. See [weaver] and [knight] for some discussion of this formula.

[wikipedia]

<http://en.wikipedia.org/wiki/Inductance>. Also see http://en.wikipedia.org/wiki/Inductance/Details_for_some_circuit_types.